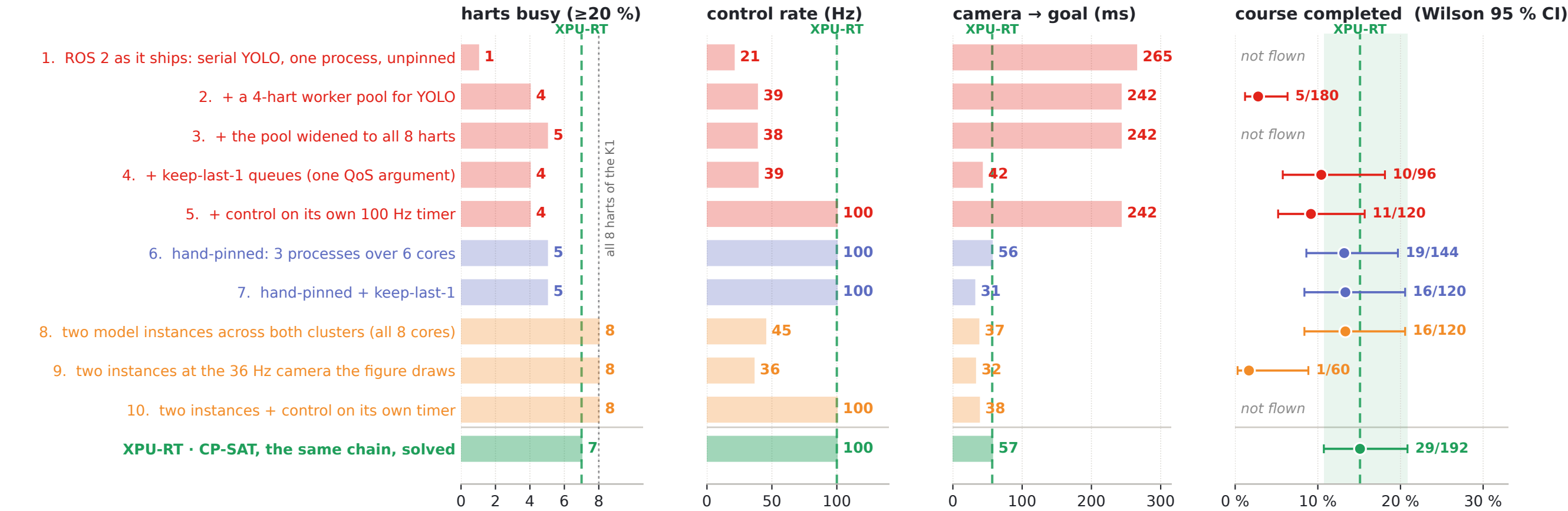


The ROS 2 baseline is a ladder, not a strawman:

each rung is one more thing an engineer does — and they climb toward a schedule that was solved

Same K1, same int8 kernels, same 45 Hz camera (rung 9 at 36 Hz). XPU-RT needs none of the rungs.

ROS 2, default placement two model instances
hand-pinned XPU-RT (reference, dashed)



Board — harts: $\geq 20\%$ busy in the per-core sampler (ROS 2 rows, median over replicates); XPU-RT at $\geq 20\%$ dispatch duty in its own board trace (8 harts carry work, 7 clear the threshold, and dispatch cycles alone are a lower bound).

Control rate = $1 / \text{mean control-output gap}$. camera $\rightarrow$ goal = median camera-to-goal chain for the ROS 2 rows, camera-to-control chain for XPU-RT.

Flights — one cell only: course a, prop density 0.30, static 2.4 m people along the aisle, deployed gain 0.0055, cadence AND latency replayed, at the 5 cruise speeds (1–1.8 m/s) and 12 seeds every drawn arm flew.

8 of 11 rows have flights in that cell; the rest are marked, never scored as zero. Rung 1's only flights replay cadence without latency, a different experiment. Pooled tallies over all campaigns: sidecar.